

LINEUM CORE // OEA

LEVEL 1

Math You Can See

*A Simple Way to
Finally Understand
Math Through Shapes*

THE OFFICIAL
DOCUMENTATION FOR FLOW
GEOMETRY



IMPRINT / LEGAL —

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CONTENTS

Part 1

The Flow of Numbers (Basic Intuition)

Part 2

Weaving Surfaces (Multiplication & Division)

Part 3

Mastering the Dimensions (Exponents & Roots)

Part 4

The Scale and The Map (Equations & Geometry)

PREFACE

Hook Intro

WHY SCHOOL MATHEMATICS FAILS YOU —

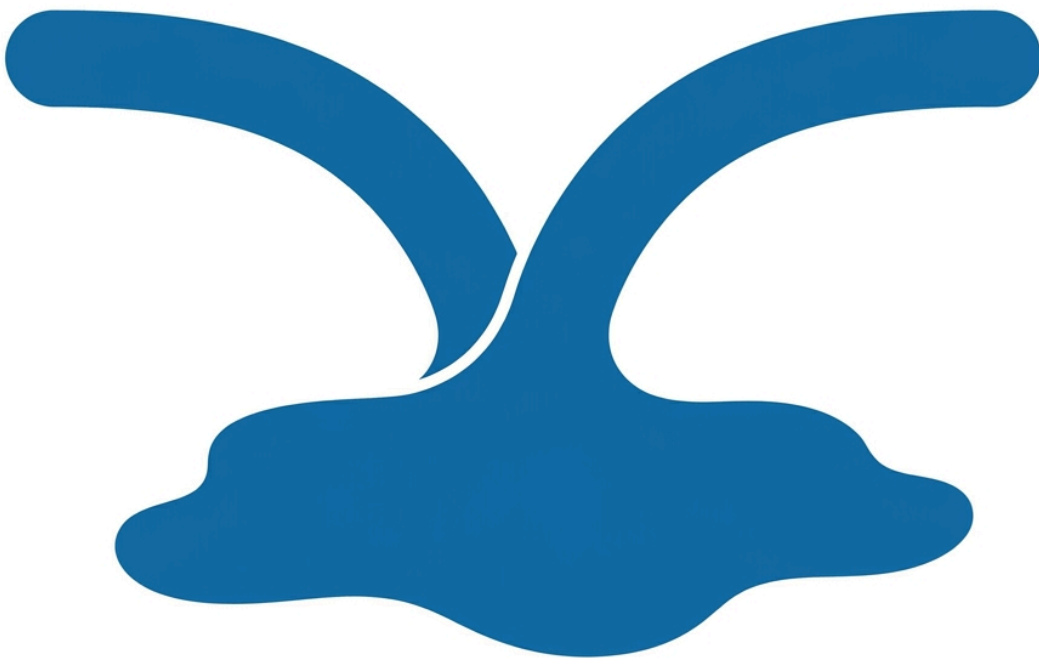
Imagine sitting in a classroom, staring at a blackboard full of abstract letters. You are told to memorize rules, move variables around, and cross out numbers. School treats math like a rigid code you must obey. But what if you could see the math instead? In this book, we translate abstract math into visual shapes. You won't memorize formulas. You will visualize blocks snapping together and structures stretching. Math makes perfect sense when you watch it happen.

HOW TO USE THIS BOOK —

Treat this book like a mental sandbox. Read the scenario at the start of each idea. Close your eyes and visualize the shape. Check the image description to anchor your imagination. Read the 'Aha Moment' to lock the concept into memory. Discover the practical steps and the pure geometry governing each rule.

Addition (+)

Imagine two small pools of water merging on a flat surface.



Addition is the act of bringing pieces together. The water combines into a single, deeper pool.



Positive numbers add size without creating conflict.

WHAT IT IS. —

Addition calculates the total amount of two separate groups.

HOW TO SOLVE. —

Line up the numbers vertically by place value. Add the columns from right to left. If a column totals more than 9, carry the extra 1 over to the next column.

WHY IT WORKS. —

When you add positive numbers, you extend a straight line in one direction. The shape grows longer.

SUMMARY

Addition adds pieces into the exact same space to build a larger shape.

Subtraction (—)

Imagine rowing a boat upstream and hitting a fast river current.

A minus sign fights your progress. You move forward, but the current pushes you backward.

WHAT IT IS. —

Subtraction calculates the difference between two numbers. It determines what remains after pieces are removed.



A minus sign pushes backward against your forward movement.

HOW TO SOLVE. —

Line the numbers up vertically. Subtract the bottom digit from the top. If the top digit is smaller, borrow 1 from the next column before subtracting.

WHY IT WORKS. —

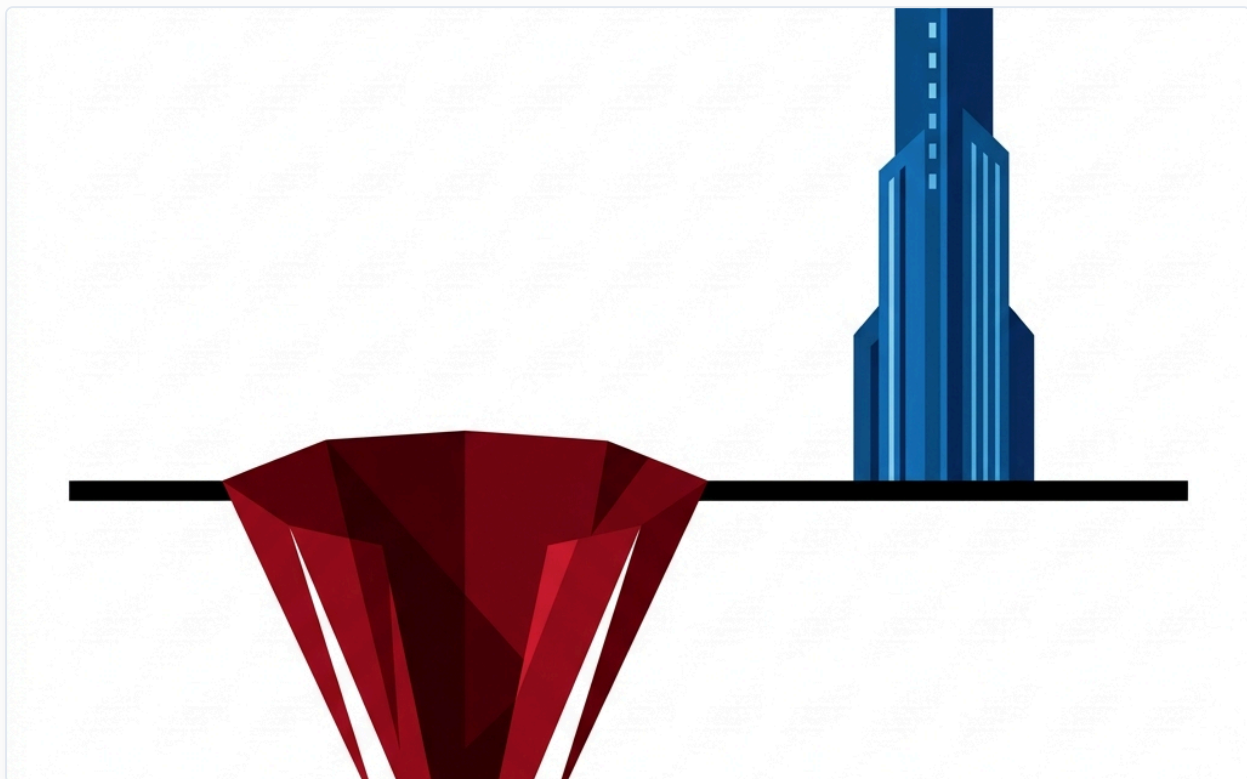
Subtraction shrinks the original shape. When you apply a minus, the shape slides backward along the grid, making it shorter.

SUMMARY

Subtracting measures opposing numbers. You calculate if your forward movement is bigger than the backward push.

Negative Numbers (Standalone)

Imagine a flat landscape. Positive numbers build towers upwards. Negative numbers dig holes downward.



A negative number represents debt or depth.



Negative numbers extend the world downward below zero.

WHAT IT IS. —

A negative number is any value less than zero. It is the exact opposite of a positive amount.

HOW TO SOLVE. —

To subtract a negative number, like $5 - (-3)$, change the double minus into a plus. The equation becomes $5 + 3 = 8$.

WHY IT WORKS. —

Removing a deep hole is the exact geometric equivalent of filling it with dirt. The double negative flips a downward hole into an upward block.

SUMMARY

Removing a hole means filling it. Two negatives flip to form a positive block.

Absolute Value ($| - 5 |$)

Imagine driving a car. Driving five miles forward consumes fuel. Driving five miles in reverse consumes the same fuel.

Absolute value ignores whether you traveled forward or backward. It focuses exclusively on the total distance.

WHAT IT IS. —

Absolute value calculates the strict distance a number sits from zero.



Absolute value is the total distance traveled, ignoring direction entirely.

HOW TO SOLVE. —

Look at the number inside the parallel bars $|-5|$. Remove the negative sign. The result is always positive or zero. $|-5|$ becomes 5.

WHY IT WORKS. —

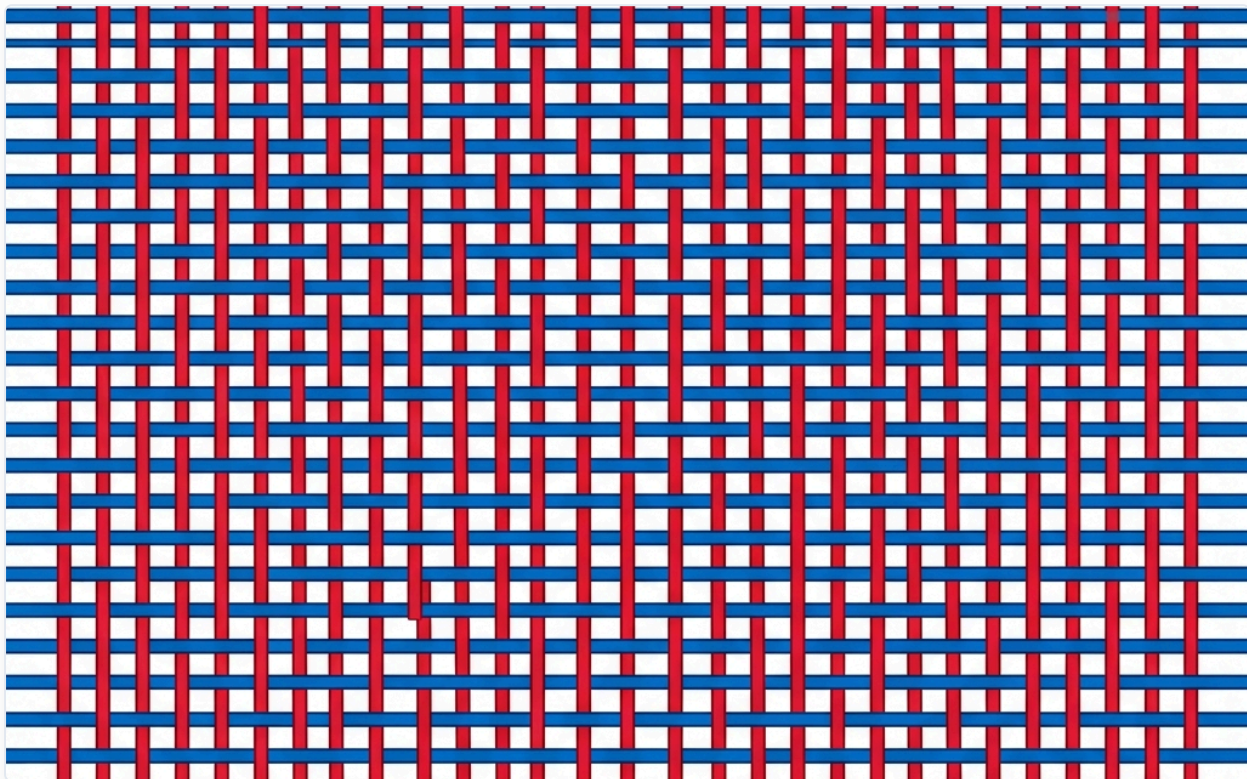
The parallel bars act as a geometric filter. They remove direction from the grid and force the object to report its total size.

SUMMARY

Absolute value removes minus signs and reports the raw size of the number.

Multiplying Variables ($x \cdot x$)

Imagine weaving horizontal blue threads crosswise with vertical red threads.



When you multiply a variable by another variable, you weave them together. You take lines and lock them into a flat 2D surface (x^2).



Multiplying variables weaves lines into solid flat shapes.

WHAT IT IS. —

Multiplying letters builds a new term with a higher exponent.

HOW TO SOLVE. —

Count the number of identical letters being multiplied. Write that number as a small exponent at the top right. For $x \cdot x$, write x^2 .

WHY IT WORKS. —

A single x is a flat line. Multiplying it by another x stretches that line sideways to make a square. Multiplying a third time stretches the square upward to form a 3D box.

SUMMARY

Multiplying variables creates shapes. Variables like x^2 act as physical objects, not abstract letters.

Multiplying by Numbers ($5 \cdot x$)

Imagine turning the volume dial on a speaker.

Standard numbers act as amplifiers. Multiplying a glowing wire by 5 makes it glow 5 times brighter. The shape of the wire stays identical.

WHAT IT IS. —

Multiplying a variable by a raw number increases the quantity of that item.



Numbers act as amplifiers, scaling the amount without altering the shape.

HOW TO SOLVE. —

When you add $2x + 3x$, you add the numbers together to get $5x$. You never change the exponent.

WHY IT WORKS. —

You are placing five individual wires side-by-side ($5x$). Because they never cross, they cannot weave into a square. The shape stays flat.

SUMMARY

Multiplying by a number ($5x$) scales the amount, while multiplying the variables (x^2) builds a completely new shape.

Division ($\div$)

Imagine breaking a large chocolate bar into equal smaller squares.



The top number is the total size. The bottom number is how many pieces you must break it into.



Division separates a total amount into identical pieces.

WHAT IT IS. —

Division splits a total amount into equal groups to see how much goes into each piece.

HOW TO SOLVE. —

To solve $20 \div 4$, ask yourself what number multiplied by 4 equals 20. The answer is 5.

WHY IT WORKS. —

Division breaks a large block into a grid of smaller exact squares. The starting shape is cut into equal pieces.

SUMMARY

Division cracks a big shape perfectly into smaller, identical pieces.

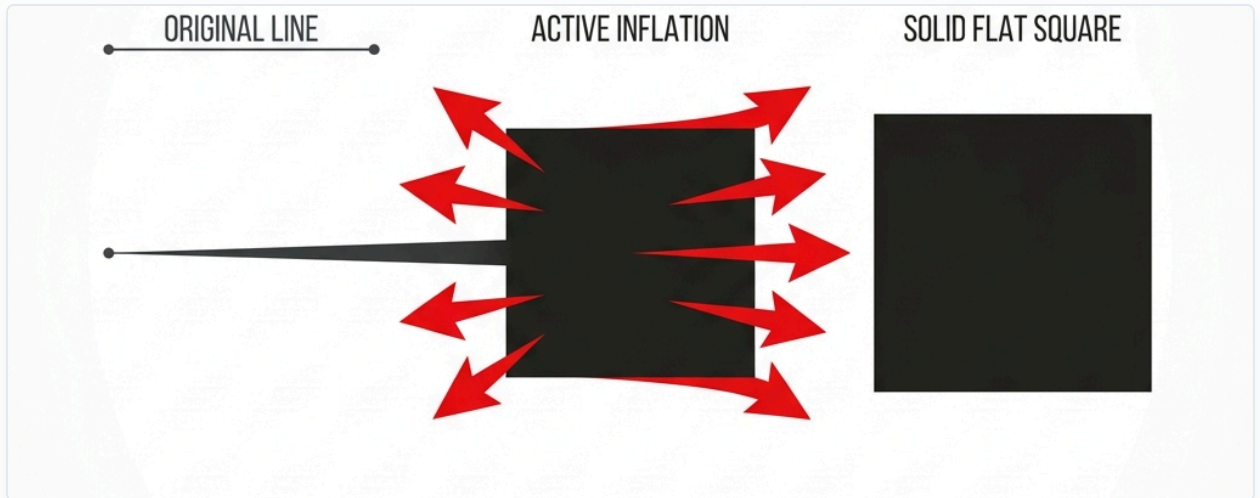
Exponents (x^2 , x^3)

Imagine a thin wooden rod inflating sideways to become a solid flat table.

Squaring an object (x^2) pulls a 1D line into a 2D plane. Cubing it (x^3) raises that plane vertically into a 3D box.

WHAT IT IS. —

An exponent tells you how many times to multiply the base number by itself.



Exponents scale objects upward into entirely new dimensions of space.

HOW TO SOLVE. —

To solve 4^3 , you multiply $4 \times 4 \times 4$. The first pair equals 16. Then multiply 16×4 to get 64.

WHY IT WORKS. —

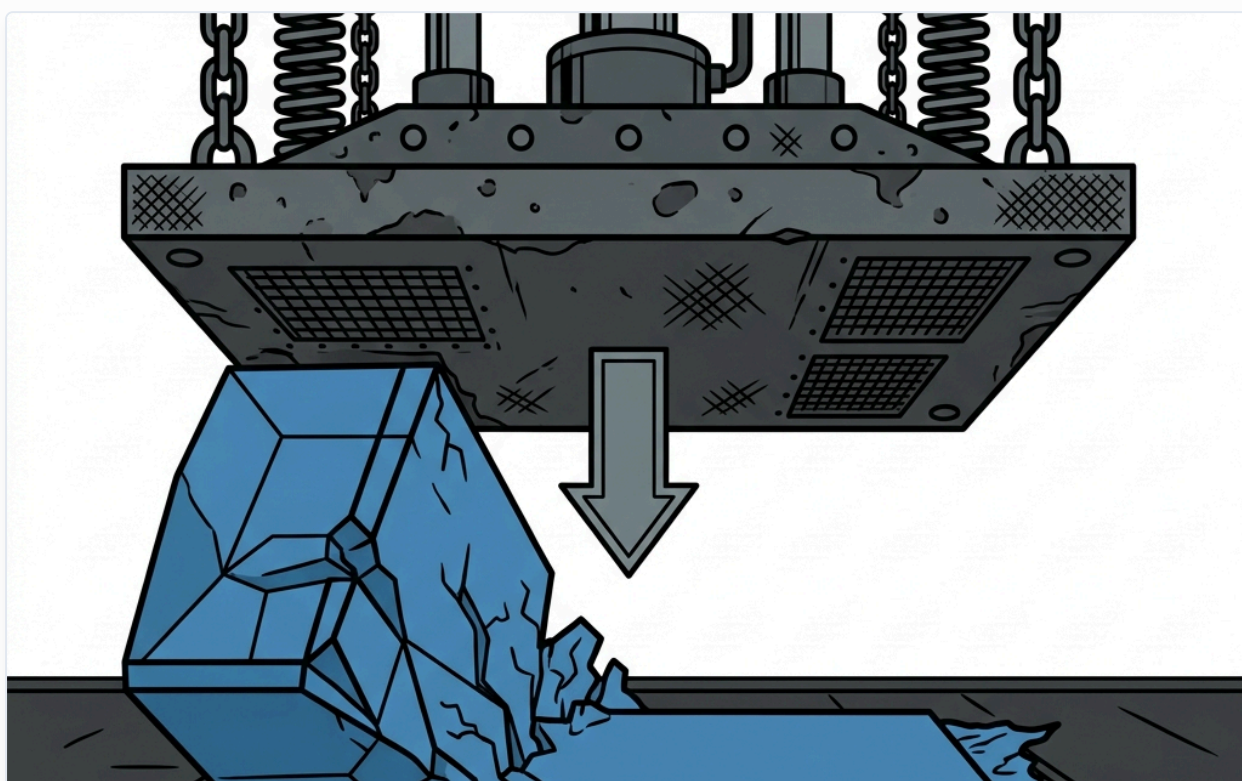
Exponents stretch shapes. You drag a line into a square, and pull that square upwards into a box.

SUMMARY

Exponents stretch shapes from lines, to squares, to boxes.

Square Roots ($\sqrt{x}$)

Imagine an industrial press flattening a solid box back down to the floor.



Roots reverse shape expansion. A square root compresses a 2D square firmly back down into a 1D line.



*Roots compress shapes
down into lower dimensions.*

WHAT IT IS. —

A square root asks what original base number was multiplied by itself to produce the current value.

HOW TO SOLVE. —

To find $\sqrt{25}$, find the number that equals 25 when multiplied by itself. Because $5 \times 5 = 25$, the answer is 5.

WHY IT WORKS. —

A root presses a shape flat downward. It finds the exact length of one side of a square and discards the rest of the shape.

SUMMARY

While exponents inflate shapes, roots compress them flat.

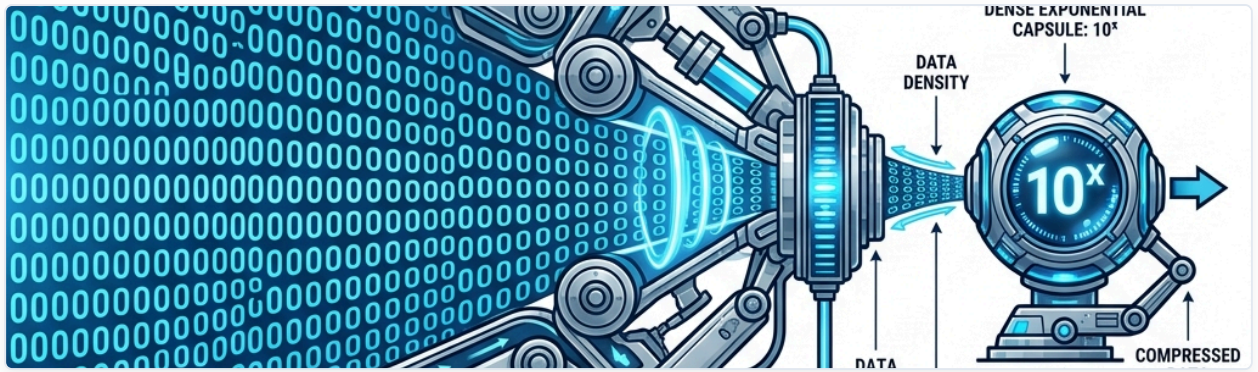
Scientific Notation ($a \times 10^b$)

Imagine packing a massive banner into a highly compressed zip file.

Writing out 300,000,000 is inefficient. Scientific notation packages all the trailing zeros into a compact format (3×10^8).

WHAT IT IS. —

It is a shorthand method for writing extremely large or extremely tiny numbers.



Scientific notation acts as simple data packing for massive numbers of zeros.

HOW TO SOLVE. —

Count the jumps from the original decimal point to the new location behind the first digit. Write that jump count as the exponent.

4, 500, 000 becomes 4.5×10^6 .

WHY IT WORKS. —

Instead of drawing a long flat line, you fold the distance onto itself into a small package that keeps the total length.

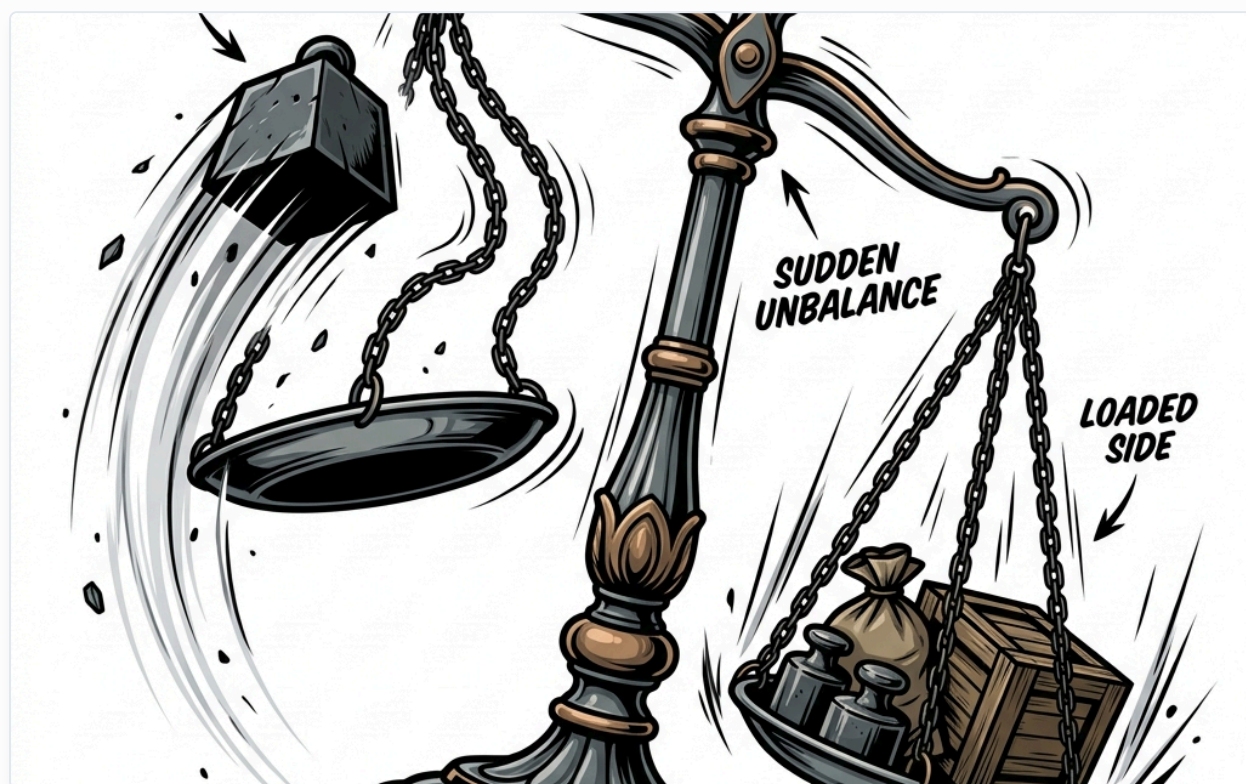
SUMMARY

Scientific notation manages massive sizes cleanly without wasting space.

Balancing Equations

$(2x + 4 = 10)$

Imagine a brass scale delicately hovering in perfect balance.



If you remove a weight from the left side, the scale tilts. To fix it, you must remove the exact same amount from the right.



Equations require you to adjust both sides identically to maintain physical balance.

WHAT IT IS. —

An equation states that the left side and the right side are exactly equal, even if they look different.

HOW TO SOLVE. —

Isolate the unknown letter. If the equation is $2x + 4 = 10$, subtract 4 from both sides to get $2x = 6$. Then divide both sides by 2 to find that $x = 3$.

WHY IT WORKS. —

A balanced shape must stay balanced. If you cut a block off the left side of a flat scale, you must cut the exact same block off the right side to keep it flat.

SUMMARY

You strip weights from both sides equally to expose the hidden value.

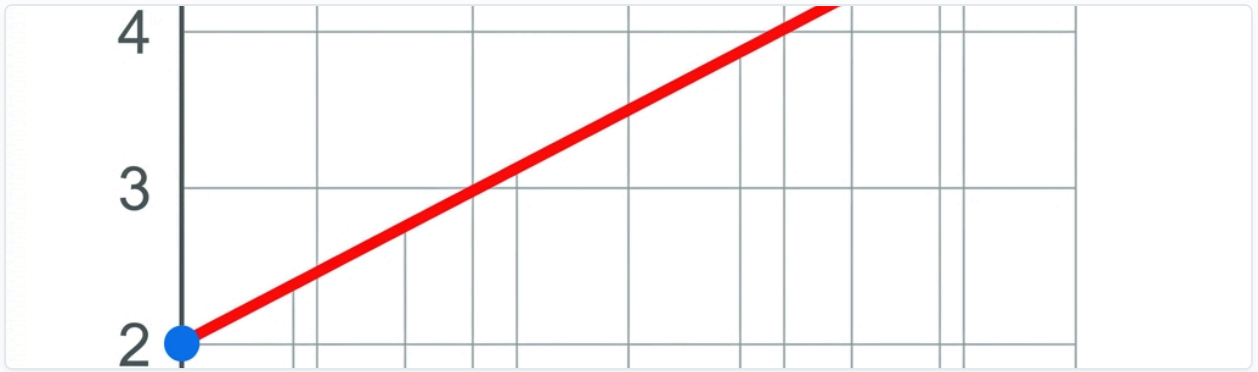
Linear Lines ($y = mx + b$)

Imagine aiming a laser pointer into the night sky from a specific step on a ladder.

The constant ' b ' is the exact step on the ladder where the laser starts.
The slope ' m ' is the angle of the beam pointing upward.

WHAT IT IS. —

It is the standard equation used to draw straight lines predictably on a graph.



A linear equation is the blueprint for a straight line.

HOW TO SOLVE. —

Mark your starting point on the vertical Y-axis using ' b '. Look at the slope ' m ' as a fraction. Move up the 'rise' number, and right the 'run' number to plot your next dot.

WHY IT WORKS. —

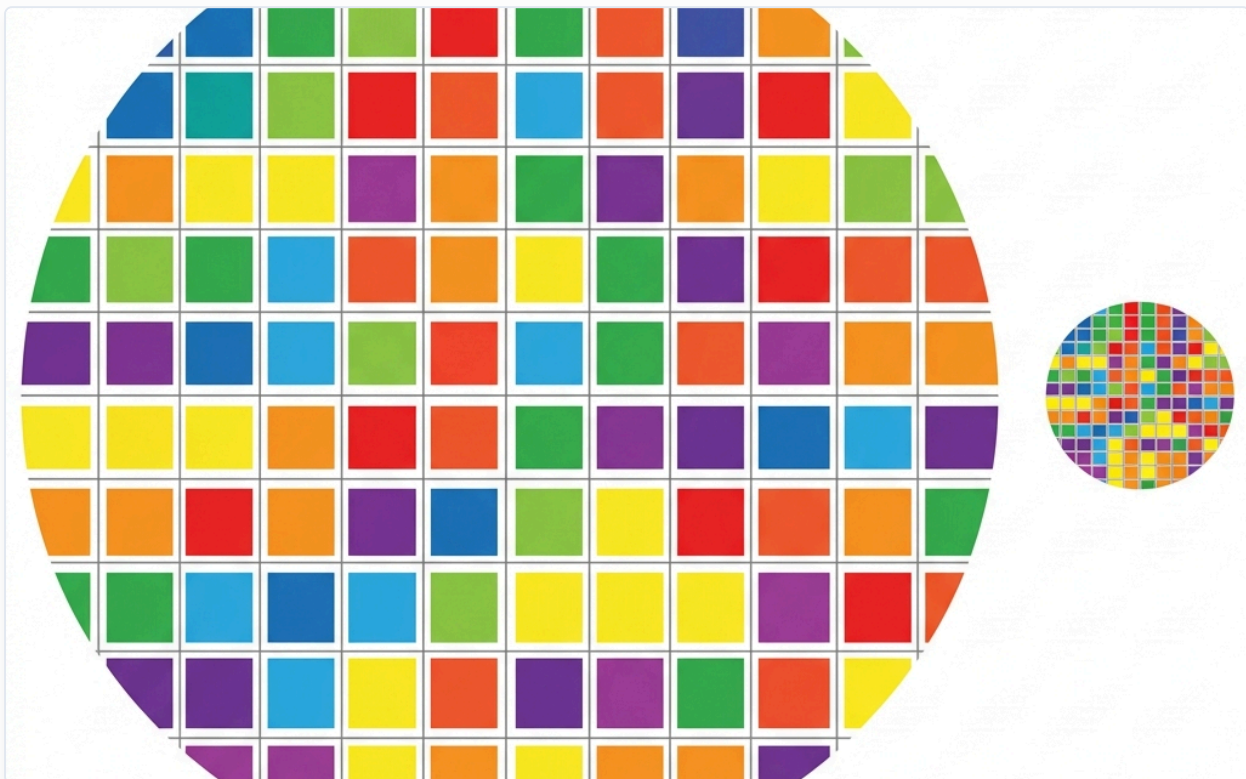
The line climbs perfectly without bending, creating identical triangles at every step.

SUMMARY

Linear lines establish a starting height, set an angle, and travel straight forever.

Percentages (%)

Imagine a transparent grid that slices any shape directly into exactly 100 equal sections.



Percentages force different objects into the exact same 100-piece scale. "50%" indicates you select exactly half of those available sections.



Percentages force completely different shapes into a universal 100-piece scale.

WHAT IT IS. —

A percentage is a ratio that compares a number specifically out of a set of 100.

HOW TO SOLVE. —

To find 30% of a number, convert the percentage into a decimal by moving the dot two places left (0.30). Then multiply the original number by 0.30.

WHY IT WORKS. —

You put a universal mask over random shapes. By chopping every object into 100 identical pieces, comparing sizes becomes easy.

SUMMARY

Percentages allow completely different sizes to share the exact same 100-piece grid.

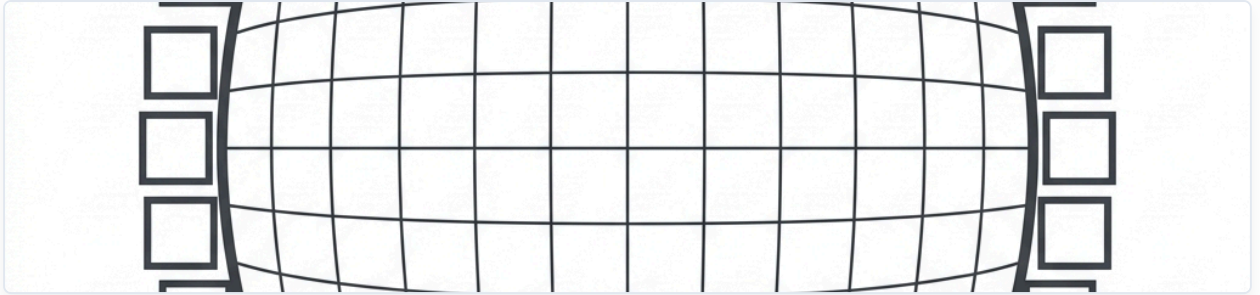
Percentage Increase/Decrease

Imagine a grid of 100 blocks expanding outward to add 20 new edge pieces.

Applying a 20% increase tells the system to attach 20 fresh blocks directly to the rim of the original shape.

WHAT IT IS. —

It calculates the exact amount a value grows or shrinks relative to its initial size.



Percentage increases add calculated modular blocks directly onto an existing structure.

HOW TO SOLVE. —

To increase 50 by 20%, multiply 50×1.20 . The '1' keeps the original value intact, while the '.20' calculates the brand new addition. The answer is 60.

WHY IT WORKS. —

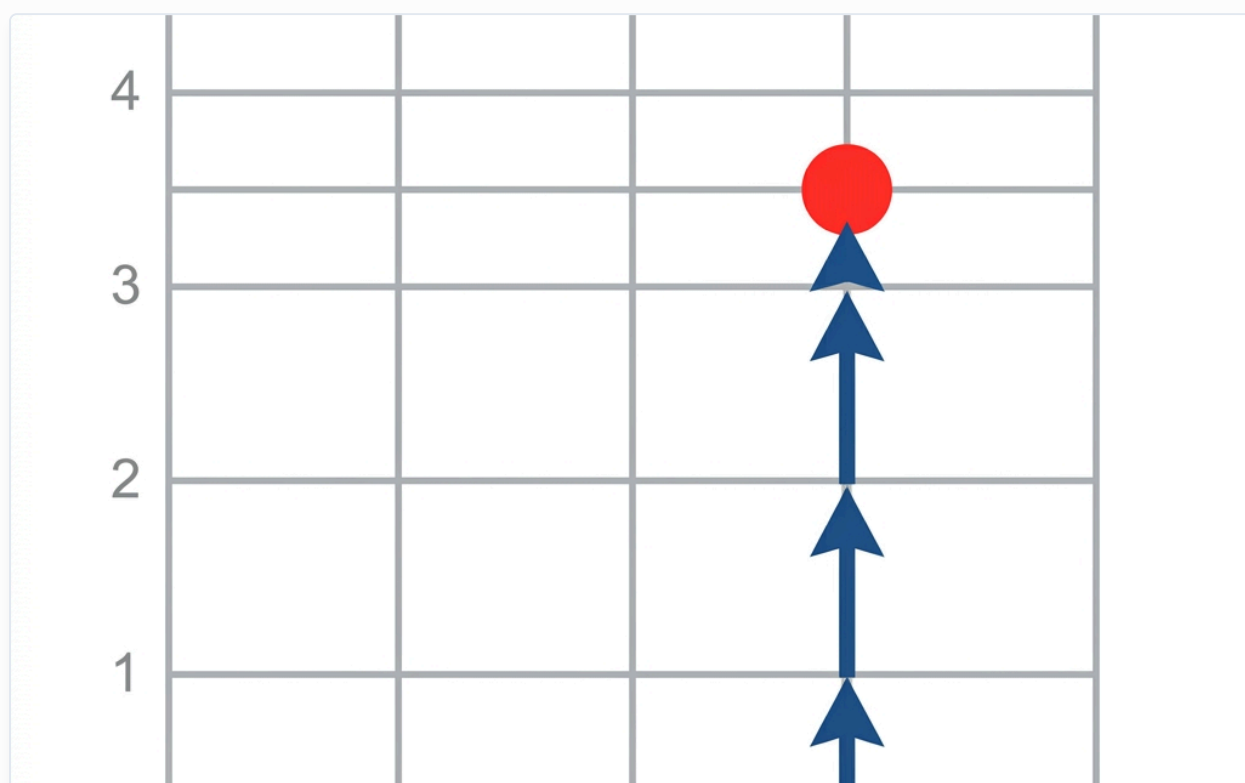
Expansion snaps pieces directly onto the outside border. The shape widens evenly.

SUMMARY

You dynamically scale the object based on how many extra 100-scale blocks fall away or attach.

The Coordinates Grid (X and Y)

Imagine walking through a city block grid using exact directions.



To find a location, you determine a starting point, walk a precise number of units right (the X axis), turn 90 degrees, and walk forward (the Y axis).



The coordinate system is a rigid map locating points perfectly.

WHAT IT IS. —

It is a 2D mapping system that uses a horizontal and vertical number line to pinpoint locations.

HOW TO SOLVE. —

Plot the point $(3, -4)$. Start at the absolute center $(0, 0)$. Move 3 units right on the X -axis. Move 4 units down on the Y -axis. Place your dot.

WHY IT WORKS. —

Coordinates pin shapes onto a flat floor. The grid stops a point from drifting away completely.

SUMMARY

Coordinates act as an irrefutable GPS map pinning geometry perfectly down.

